

Course Syllabus

Graphical Models

- Lecture 1: Introduction and Bayesian Network

Inference

- Lecture 2: Variational Inference
- Lecture 3: Variational Autoencoder (VAE)

Parameter Learning

- Lecture 4: Maximum Likelihood and EM Algorithm

Deep Generative Models

- *Generative Adversarial Network (GAN)*

- Lecture 5: Introduction to GAN

- *Large Language Model (LLM)*

- Lecture 6: Introduction to LLM
- Lecture 7: Multi-modal LLM and Parameter-Efficient Finetuning
- Lecture 8: LLM Reasoning
- Lecture 9: LLM Agent

- *Diffusion Model*

- Lecture 10: Denoising Diffusion Model
- Lecture 11: Score-based Diffusion Model

Implementation of Deep Generative Models

- Lecture 12 (Guest Lecture): Implementation of Variational Autoencoder
- Lecture 13 (Guest Lecture): Implementation of Generative Adversarial Network
- Lecture 14 (Guest Lecture): Implementation of LLM (I): Transformer and language model
- Lecture 15 (Guest Lecture): Implementation of LLM (II): Multi-modal LLM and parameter-efficient finetuning
- Lecture 16 (Guest Lecture): Implementation of Denoising Diffusion Model
- Lecture 17 (Guest Lecture): Implementation of Score-based Diffusion Model

Advanced Topics and Course Project Presentation

- Lecture 18: World Models
- Lecture 19: Course Project Presentation (I)
- Lecture 20: Course Project Presentation (II)

Grade breakdown

1. Three homework (30%)

- Each homework 10%

2. One course project (70%)

- Project proposal (10%)
- Mid-term report (20%)
- Final report (40%)

Assignment due dates

The due time is 11:59 pm PT for all dates.

Project proposal: Jan 28

Homework 1: Feb 2

Homework 2: Feb 16

Project mid-term report: Feb 18

Homework 3: Mar 2

Project final report: Mar 18